Ⓑ 2025-J119-1-EL-01/20

Pre-Junior Cycle Final Examination, 2025

Mathematics

Ordinary Level

Time: 2 hours

270 marks

CANDIDATE DETAILS

DAY and MONTH of BIRTH	/	For example, 3rd February is entered as 03/02
NAME		
SCHOOL		
TEACHER		

School Stamp

For Examiner only	
Question	Mark
1.	
2.	
3.	
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5.	
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9.	
10.	
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13.	
Total	
%	

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Ⓑ 2025-J119-1-EL-01/20

Instructions

There are 13 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. You may ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1 **(Suggested maximum time: 5 minutes)**

(a) Find the value of each of the following.

(i) $219 + 367$

(ii) $9{\cdot}2 \times 7$

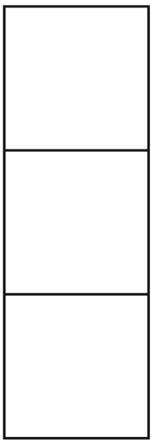
(iii) $6^2 - (9 \div 3)$

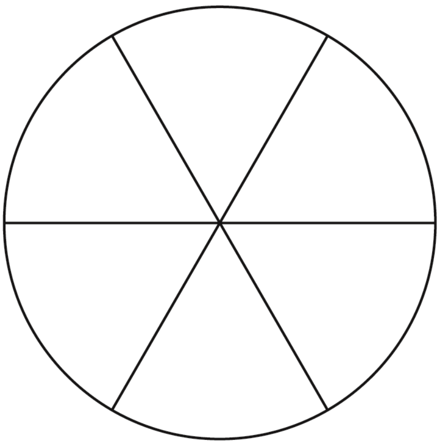
(b) Write the following numbers in order, from the largest to the smallest.

$1{\cdot}3, \quad \sqrt{2}, \quad \dfrac{2}{3}$

Answer = ☐ , ☐ , ☐ .

(c) Shade in $\frac{2}{3}$ of the area of each shape below.





Question 2 **(Suggested maximum time: 15 minutes)**

The height of all the players on the panel of a school hurling team are recorded below.

173	165	183	168	181
158	161	163	149	164
172	177	179	165	179
171	175	169	171	172
163	159	163	181	164

(a) In **total**, how many students are on the panel?

Answer =

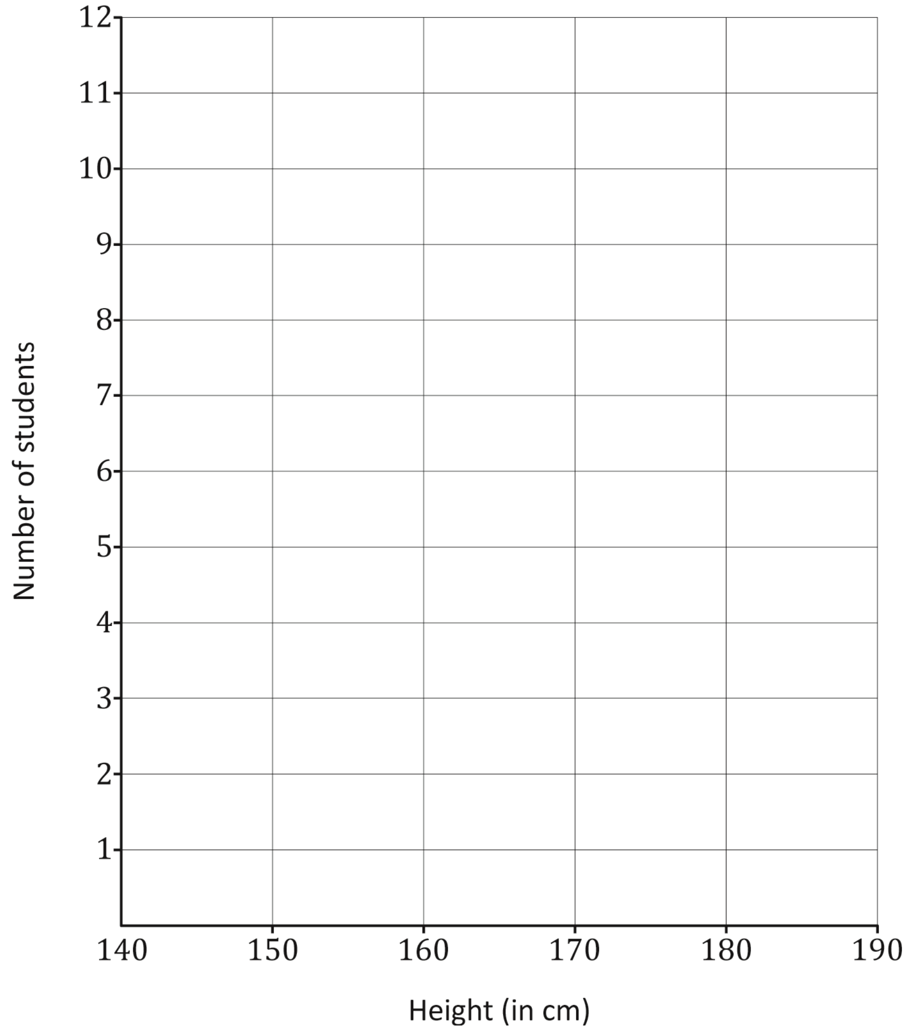
(b) Find the **range** of the heights of the students on the hurling panel.

(c) Find the **median** height of the students on the hurling panel.

(d) Complete the following frequency table.

Height (in cm)	**140 – 150**	**150 – 160**	**160 – 170**	**170 – 180**	**180 – 190**
Number of students					

(e) Draw a **histogram** to represent the data from the frequency table.
Use the axes and scales below.



(f) Write down the **modal height** interval of the students on the hurling panel.

Modal height interval =

Question 3 **(Suggested maximum time: 10 minutes)**

(a) Filip buys the following lunch items:

				
Fresh milk	Yogurt	Fresh Soup	Apples	Bread
€0·85	€1·15	€2·89	€2·20	€2·19

(i) Work out the **total cost** of these items.

(ii) Filip pays for his lunch items with a €20 note.
Work out how much **change** he receives.

(b) VAT is charged at a rate of 23% on sweets and chocolates in Ireland.
A packet of sweets costs €2·44 before VAT is added.
Work out the **total cost** of this packet of sweets, including the VAT.
Give your answer correct to the nearest cent.

(c) Another brand of sweets cost €2·85 for a 150g packet.
Work out the **cost per kg** of this packet of sweets.
Give your answer correct to the nearest cent.

Question 4 (Suggested maximum time: 10 minutes)

(a) Find the value of $4a^2 + 3b$, when $a = 2$ and $b = -2$.

(b) Solve the following equation in x:

$$3x - 6 = 18$$

(c) Factorise $x^2 - 144$.

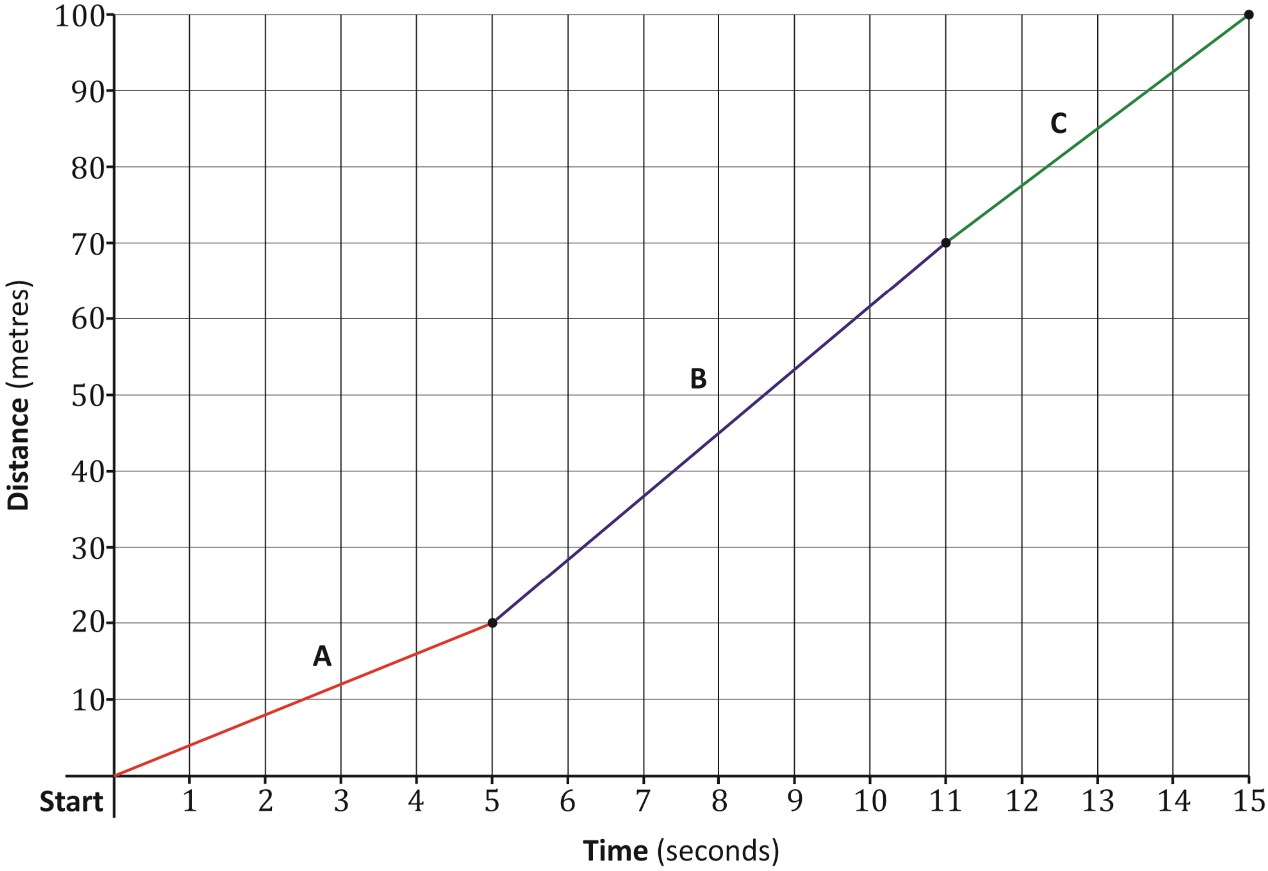
(d) Factorise fully $rc - sc + 2rd - 2sd$.

Question 5 **(Suggested maximum time: 5 minutes)**

Peter took part in a 100 m race.

The following graph shows the distance, in metres, ran by Peter after t seconds during the race.

The graph shows three stages labelled **A**, **B** and **C**.



(a) How many seconds did it take Peter to finish the race? Answer = sec

(b) What distance had Peter travelled after 11 seconds? Answer = m

(c) Tick the correct box to show what section of the graph indicates Peter's slowest average speed during the race.

Tick (✓) **one** box only.
Give a reason for your answer.

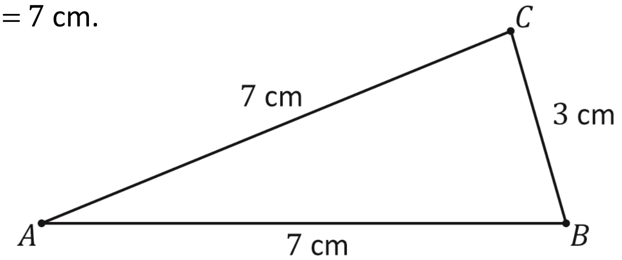
A	**B**	**C**
☐	☐	☐

Reason:

(d) Work out Peter's **average speed**, in metres per second, during the stage labelled **A** in the diagram.

Question 6 **(Suggested maximum time: 10 minutes)**

A sketch of the triangle ABC is shown (not to scale).
$|AB| = 7$ cm, $|BC| = 3$ cm and $|AC| = 7$ cm.



(a) Tick the correct box to show what type of triangle this is.

Tick (✓) **one** box only. Give a reason for your answer.

Equilateral	Right-angled	Isosceles
☐	☐	☐

Reason:

(b) Construct the triangle ABC.
Each small square in the grid has sides of length 1 cm. The point A is marked for you.
Show your construction lines clearly.

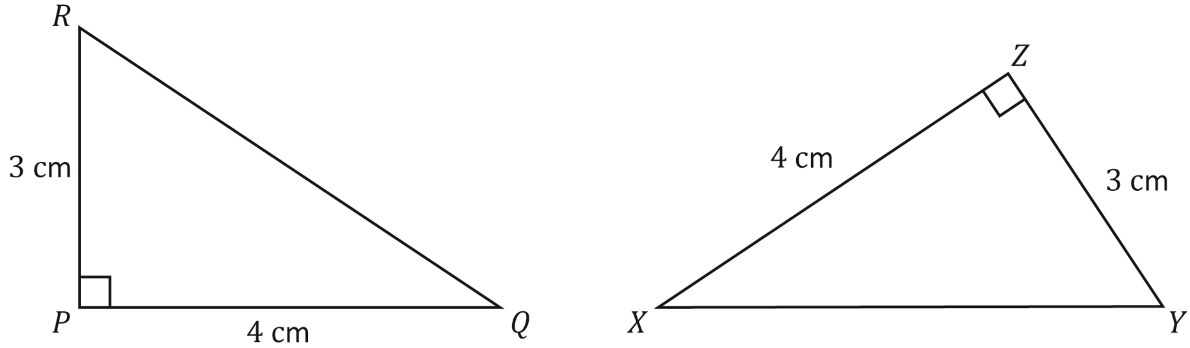
A

(c) Measure the size of the smallest angle in your construction above.
Write the size of the angle in the box below.

Size of the smallest angle = $\boxed{\quad}^\circ$

Question 7 **(Suggested maximum time: 5 minutes)**

Two right-angled triangles PQR and XYZ are shown below (not to scale).



(a) Use the theorem of **Pythagoras** to work out the length, $|QR|$.

(b) Fill in the missing statements and reasons into the table below to show that the triangles PQR and XYZ are **congruent**.

Statement	Reason
Side: $\quad |PQ| = |XY|$.	
Angle:	Both angles are 90°.
Side:	Both sides are 3 cm in length.

(c) Write down the length, $|XY|$.

$|XY| =$ ☐ cm

Question 8 **(Suggested maximum time: 5 minutes)**

Footgolf is combination of football and golf.
The game is played in the same way as golf, except players use a football instead of a golf ball, and the ball is kicked rather than struck with a club.

John, Emer and Keith are playing footgolf.
John takes x shots to complete the first hole.

(a) Emer takes one more shot than John to complete the first hole.
Tick the correct box to show what expression represents the number of shots Emer took.

Tick (✓) **one** box only.

$x+1$ ☐ $x-1$ ☐ $1x$ ☐

(b) Keith takes twice as many shots as John to complete the first hole.
Tick the correct box to show what expression represents the number of shots Keith took.

Tick (✓) **one** box only.

$x+2$ ☐ $x-2$ ☐ $2x$ ☐

(c) It took John, Emer and Keith 17 shots in total to complete the hole.
Use this information to work out the value of x, the number of shots it took John to complete the hole.

(d) John's scores for the last four holes are shown below.

2	3	4	5

Tick the correct box to show what kind of sequence is made by these scores.

Tick (✓) **one** box only. Give a reason for your answer.

linear ☐ non-linear ☐

Reason:

Question 9 **(Suggested maximum time: 5 minutes)**

(a) The attendance at the 2024 GAA All-Ireland Senior Football final was 82 274.

(i) Write 82 274 correct to **2 significant figures**.

Answer =

(ii) Sonya rounded 82 274 correct to the **nearest 100**.
Write Sonya's estimate in the answer box below.

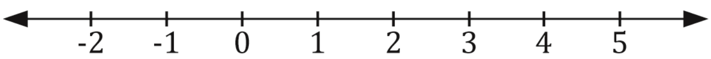
Answer =

(iii) Write Sonya's estimate in the form $a \times 10^n$, where $1 \le a < 10$ and $n \in \mathbb{N}$.

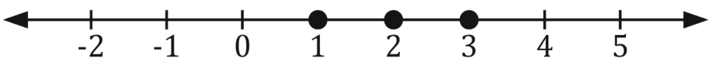
(b) Fill in the table below to show each number in the form 10^n where $n \in \mathbb{N}$.
One has already been done for you.

Number	**Number in the form 10^n**
100	
$10 \times 10 \times 10 \times 10$	10^4
$10^4 \times 10^3$	
$(10^2)^3$	

(c) **Graph** the inequality, $x < 1$ where $x \in \mathbb{R}$, on the numberline below.



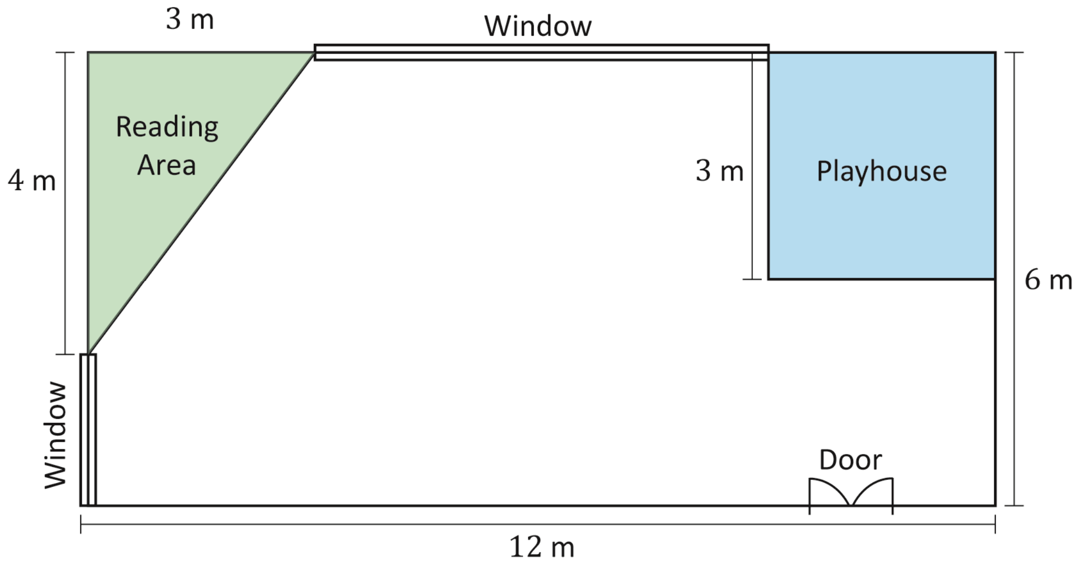
(d) Tick the correct box to show which inequality is illustrated on the numberline below.
Tick (✓) **one** box only.



$x \le 3,\ x \in \mathbb{Z}$	$x < 3,\ x \in \mathbb{N}$	$x \le 3,\ x \in \mathbb{N}$
☐	☐	☐

Question 10 **(Suggested maximum time: 15 minutes)**

Alex is renovating the playroom of her home. The playroom measures $12 \text{ m} \times 6 \text{ m}$.
She has divided the room into different areas according to activities as shown below.



(a) Work out the total area of the playroom.

(b) The reading area is in the shape of a right-angled triangle.
Work out the area of this section of the room.

(c) The area of the playhouse is 9 m^2.
Work out the width of the playhouse.

(d) Alex buys a circular rug as a play mat for 'circle time' in the playroom.
The diameter of the rug is 4 m.

(i) Write down the length of the radius of the rug.

Radius = m

(ii) Use the following formula to work out the area of the rug.
Give your answer in m^2, correct to 1 decimal place.

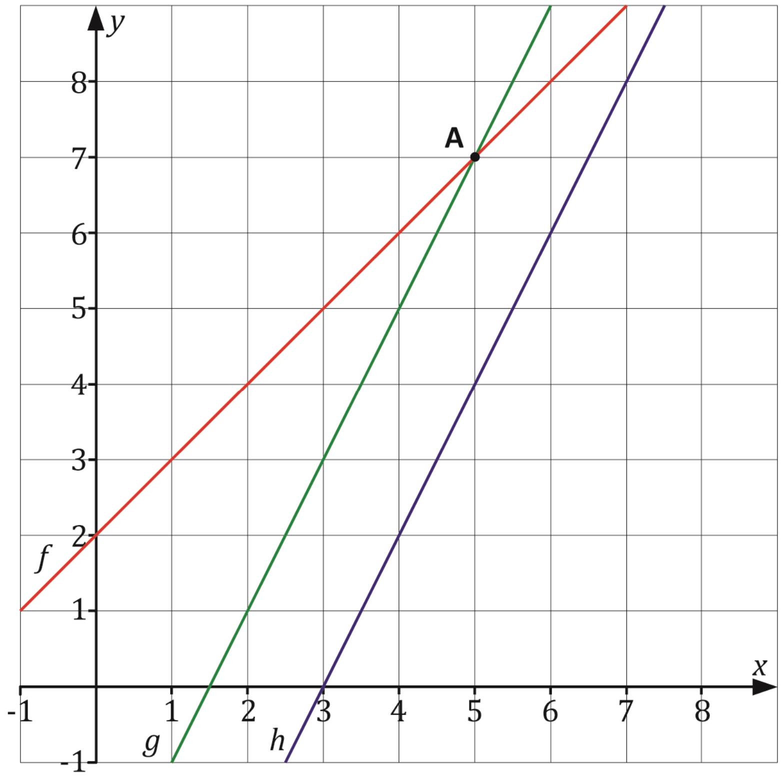
$A = \pi r^2$

(e) Peter tells Alex that the rug will not fit in the room.
Show that Peter is incorrect by indicating in the diagram a suitable place where the rug would fit in the playroom.

(f) Work out the area of the floor **remaining** when the reading area, playhouse and the 'circle time' rug are put in place.
Give your answer in m^2, correct to 1 decimal place.

Question 11 **(Suggested maximum time: 10 minutes)**

The graph on the co-ordinate diagram below shows the lines f, g and h.



(a) The point **A** is marked on the graph. Write down the co-ordinates of the point **A**.

Answer = (,)

(b) The slope of the line f is 1.
Work out the equation of the line f, giving your answer in the form $y = mx + c$.

(c) Write down the co-ordinates of the point where f intersects the y-axis.

Answer = (,)

(d) Work out the slopes of the line g and the line h.

slope of g =

slope of h =

(e) Tick the correct box to show the relationship between the line g and the line h.

Tick (✓) **one** box only. Give a reason for your answer.

parallel ☐ perpendicular ☐

Reason:

Question 12 **(Suggested maximum time: 5 minutes)**

A shop sells a tray of Pringles Crisps for €9·75.
The shop buys them for €7·00 from the wholesaler.

(a) The shop buys 50 trays of Pringles Crisps.

(i) How much profit will the shop make if it sells all 50 trays?

(ii) Write the shop's profit on the sale of Pringles Crisps as a **percentage** of its cost price.

$$\% \text{ Profit} = \frac{\text{Profit}}{\text{Cost price}} \times 100$$

(b) Each tub of Pringles Crisps is in the shape of a cylinder.
It has the dimensions as shown in the diagram.

Work out the volume of each of the tubs.
Give your answer correct to the nearest cm^3.

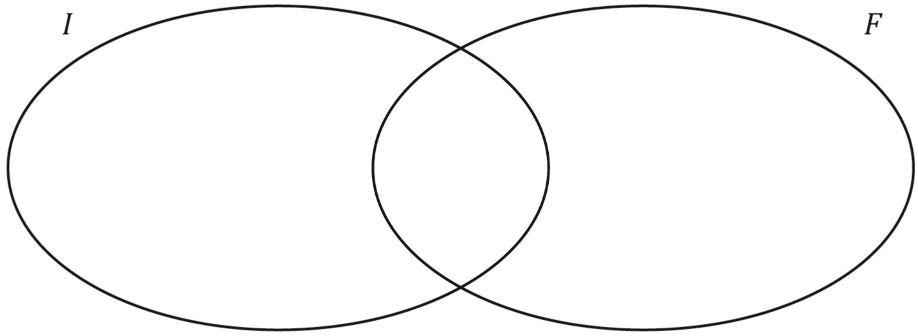
$$V = \pi r^2 h$$

Question 13 **(Suggested maximum time: 5 minutes)**

A group of students were asked if they are studying Irish (I) or French (F).
All students study at least one of the two languages.

The probability of a student studies both Irish and French is $0{\cdot}3$.
The probability of a student studies Irish only is $0{\cdot}6$.

(a) Fill in the Venn diagram to show the above information.



(b) Write down the probability that a student studies French only.

(c) Tick the correct box to show the probability that a student studies Irish.

Tick (✓) **one** box only.

$0{\cdot}1$	$0{\cdot}6$	$0{\cdot}9$
☐	☐	☐

Page for extra work.
Label any extra work clearly with the question number and part.

Q. & part	

Pre-Junior Cycle Final Examination, 2025

Mathematics – Ordinary Level

Time: 2 hours

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